

Chaoyi Wang

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I am a Ph.D. student at the University of Chinese Academy of Sciences, conducting research at the Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences. I received my M.S. degree from Johns Hopkins University and my B.S. degree from Shanghai Jiao Tong University. My research focuses on **embodied artificial intelligence**, especially embodied perception, multimodal reasoning, robust sensing, and decision-making under partial observation. I am interested in building intelligent agents that can perceive the physical world, reason over incomplete visual and sensory evidence, and make reliable decisions in real-world environments.

My current research is centered on **embodied multimodal intelligence**. A key theme of my work is how an agent can understand the world when visual evidence is incomplete, noisy, occluded, or ambiguous. In the OCC-MLLM series, I study occlusion reasoning with multimodal large language models, 3D-aware supervision, chain-of-thought guidance, self-correction, and joint understanding-generation mechanisms. Rather than treating recognition as a closed-set classification problem, these works explore how multimodal agents can infer hidden object properties from partial observations, commonsense priors, visual context, and structured reasoning. This direction is closely connected to embodied AI, where agents often need to act based on incomplete perception, limited viewpoints, and uncertain observations.

Another important direction of my research is **UAV perception and sensor-based intelligence**. My earlier work investigated real-time vehicle sound detection, multichannel acoustic moving vehicle detection, radar-based multi-target detection and tracking, low-resolution recognition, infrared-visible image fusion, and small object detection. These projects provided me with a strong background in signal processing, robust perception, and sensing systems under real-world constraints. I am particularly interested in connecting these sensing foundations with embodied agents, especially UAV systems that must perceive, localize, identify, and reason under limited onboard computation, domain shift, noisy signals, and constrained viewpoints.

In parallel, I also work on **generative artificial intelligence and video intelligence**. My recent projects include video subtitle removal with end-to-end mask-free inference, consistency tuning for video generation, adaptive moving dubbing benchmarks for film-making production, whole-body digital human generation, multi-conditional text-to-image generation, and reasoning-guided image editing. These works reflect my broader interest in controllable multimodal generation, temporal consistency, visual reasoning, and practical AI systems for creative and production workflows. While this line is related to generative AI, I see it as complementary to my embodied AI research: both require models to understand visual structure, maintain consistency, and reason beyond surface-level perception.

Overall, my research lies at the intersection of **embodied AI, multimodal large language models, sensor intelligence, and generative vision**. I aim to develop AI systems that are not only capable of recognizing patterns, but also able to reason about physical scenes, handle uncertainty, integrate multimodal evidence, and operate reliably in complex real-world environments.

As of my current Google Scholar profile, my publications have received over 80 citations, with an h-index of 5 and an i10-index of 2. My research has appeared in journals and venues including Remote Sensing, Neurocomputing, IEEE Internet of Things Journal, IEEE Transactions on Instrumentation and

Measurement, IEEE Sensors Journal, Image and Vision Computing, Computer Vision and Image Understanding, Journal of Visual Communication and Image Representation, and arXiv preprints.

Research Interests

- Embodied Artificial Intelligence
- Embodied Perception and Reasoning
- Multimodal Large Language Models
- Occlusion Reasoning and Partial Observation
- UAV Perception and Autonomous Systems
- Sensor Intelligence and Signal Processing
- Generative AI and Video Generation
- Vision-Language Reasoning

Selected Publications

Embodied AI, Multimodal Reasoning, and Occlusion Understanding

OCC-MLLM-V2: Joint understanding and generation for occluded objects via multi-modal token learning

Chaoyi Wang, Qingdong He, Y. Yang, J. Pei, L. Xia, J. Liu, B. Li, X. Di
Journal of Visual Communication and Image Representation, 2026.

OCC-MLLM-V1: Occlusion reasoning with commonsense-guided Multi-modal LLM based agent via internal Chain-of-Thoughts guidance

Chaoyi Wang, Qingdong He, J. Pei, L. Xia, J. Liu, B. Li, X. Di
Computer Vision and Image Understanding, 2026.

OCC-MLLM-CoT: Self-correction enhanced occlusion recognition with large language models via 3D-aware supervision and chain-of-thoughts guidance

Chaoyi Wang, F. Meng, J. Pei, L. Xia, J. Liu, X. Yuan, X. Di
Image and Vision Computing, 2025.

OCC-MLLM-CoT-Alpha: Towards Multi-stage Occlusion Recognition Based on Large Language Models via 3D-Aware Supervision and Chain-of-Thoughts Guidance

Chaoyi Wang, B. Li, X. Di
arXiv preprint arXiv:2504.04781, 2025.

OCC-MLLM-V1: Commonsense-Guided Multi-Modal LLM Based Agent for Occlusion Reasoning With Internal Chain-of-Thoughts Guidance

Chaoyi Wang, Qingdong He, J. Pei, L. Xia, J. Liu, B. Li, X. Di
SSRN preprint, 2025.

Sensor Intelligence, Signal Processing, and Robust Perception

Real-time vehicle sound detection system based on depthwise separable convolution neural network and spectrogram augmentation

Chaoyi Wang, Y. Song, H. Liu, H. Liu, J. Liu, B. Li, X. Yuan
Remote Sensing, 14(19), 4848, 2022.

Detection of moving vehicles from multichannel acoustic signals using convolutional recurrent neural networks

H. Cao, H. Liu, Chaoyi Wang

IEEE Sensors Journal, 24(20), 32772–32781, 2024.

HDT Network: A high-resolution range profile multi-target detection and tracking method based on neural network

L. Sun, H. Liu, Chaoyi Wang, B. Li

IET Radar, Sonar & Navigation, 17(9), 1430–1440, 2023.

B2mfuse: A bi-branch multiscale infrared and visible image fusion network based on joint semantics injection

F. Meng, X. Chen, H. Tang, Chaoyi Wang, G. Tong

IEEE Transactions on Instrumentation and Measurement, 73, 1–17, 2024.

Small object detection based on hierarchical attention mechanism and multi-scale separable detection

Y. Zhang, J. Yu, Y. Wang, S. Tang, H. Li, Z. Xin, Chaoyi Wang, Z. Zhao

IET Image Processing, 17(14), 3986–3999, 2023.

A transformer-based low-resolution face recognition method via on-and-offline knowledge distillation

Y. Song, H. Tang, F. Meng, Chaoyi Wang, M. Wu, Z. Shu, G. Tong

Neurocomputing, 509, 193–205, 2022.

LLRFaceFormer: Lightweight face transformer for real-world low-resolution recognition

Y. Song, Chaoyi Wang, H. Tang, S. Han, M. Li, G. Tong

IEEE Internet of Things Journal, 11(9), 16769–16787, 2024.

Generative AI, Video Generation, and Visual Editing

CLEAR: Context-Aware Learning with End-to-End Mask-Free Inference for Adaptive Video Subtitle Removal

Qingdong He, Chaoyi Wang, P. Tang, Y. Yang, X. Hu

arXiv preprint arXiv:2603.21901, 2026.

Preview WB-DH: Towards Whole Body Digital Human Bench for the Generation of Whole-body Talking Avatar Videos

Chaoyi Wang, Y. Yang, J. Pei, L. Xia, J. Liu, X. Yuan, X. Di

arXiv preprint arXiv:2508.08891, 2025.

Towards Film-Making Production Dialogue, Narration, Monologue Adaptive Moving Dubbing Benchmarks

Chaoyi Wang, J. Zheng, Z. Chen, S. Xia, C. Ding, X. Zhang, X. Tao, X. He, X. Di

arXiv preprint arXiv:2505.01450, 2025.

Video Generation with Consistency Tuning

Chaoyi Wang, Y. Song, Y. Zhang, J. Pei, L. Xia, J. Liu

arXiv preprint arXiv:2403.06356, 2024.

Pixelponder: Dynamic Patch Adaptation for Enhanced Multi-Conditional Text-to-Image Generation

Y. Pan, Qingdong He, Z. Jiang, P. Xu, Chaoyi Wang, J. Peng, H. Wang, Y. Cao, Z. Gan, et al.
arXiv preprint arXiv:2503.06684, 2025.

Reasoning to Edit: Hypothetical Instruction-Based Image Editing with Visual Reasoning

Qingdong He, X. Chen, Chaoyi Wang, Y. Pan, X. Hu, Z. Gan, Y. Wang, C. Wang, X. Li, et al.
arXiv preprint arXiv:2507.01908, 2025.

Short Bio

Chaoyi Wang is a Ph.D. student at the University of Chinese Academy of Sciences. He received his M.S. degree from Johns Hopkins University and his B.S. degree from Shanghai Jiao Tong University. His research focuses on embodied artificial intelligence, multimodal reasoning, sensor intelligence, and generative vision. His recent work explores occlusion reasoning with multimodal large language models, UAV-oriented perception, robust sensing, video generation, and visual editing. He is interested in building intelligent agents that can perceive, reason, and make decisions under incomplete observations and real-world uncertainty.